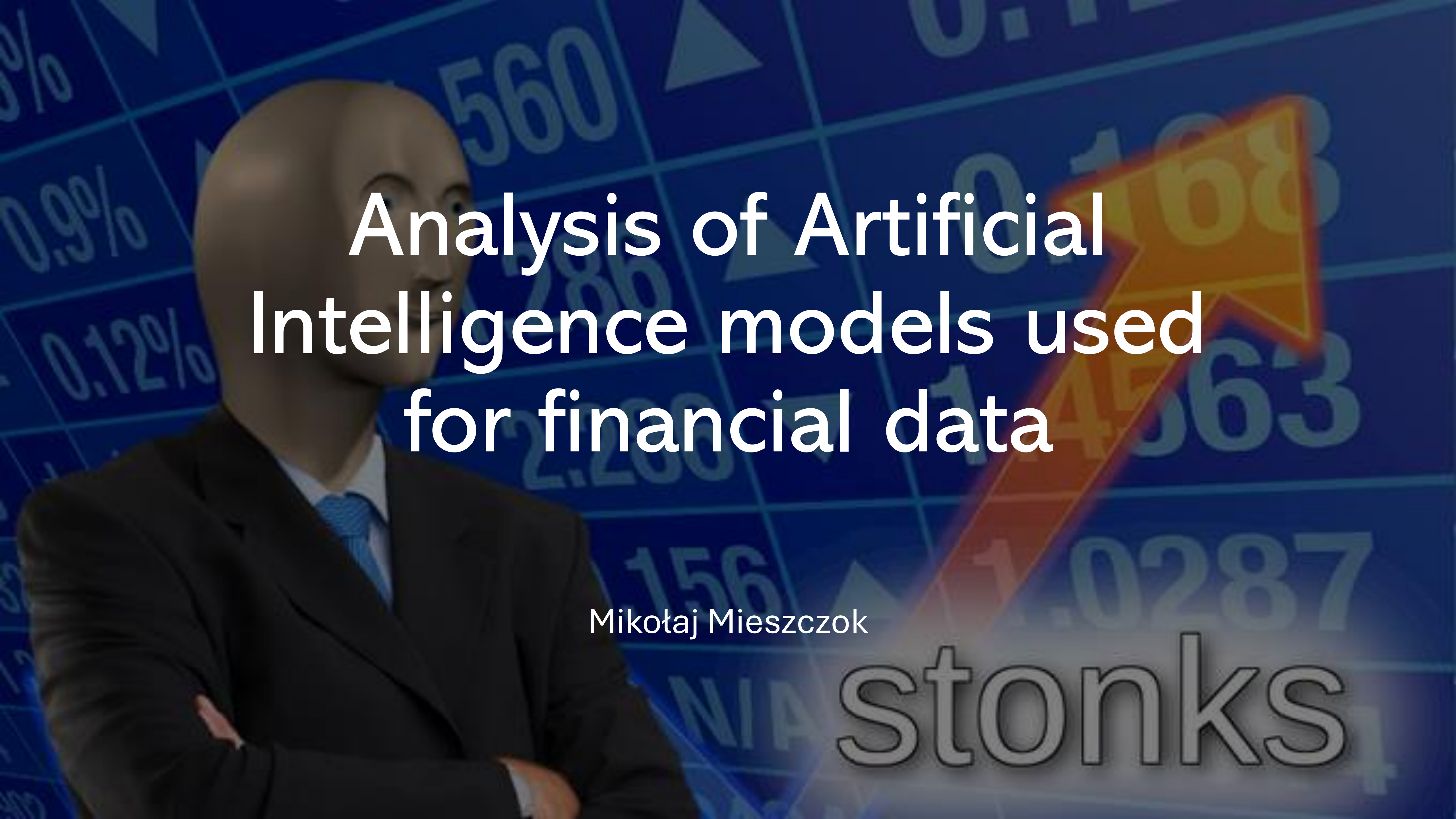


Analysis of Artificial Intelligence models used for financial data

Mikołaj Mieszczok

Motivation behind the project

Stock market is a devious playing field. The only ways to generate risk-free profit is either:

- Have a very large capital
- Have access to insider information
- Have connections in the government

For an average person the chance of having access to even one of those methods is very low, not to mention one of them is illegal. Unfortunately, or fortunately for us, the law does not apply to the rich and powerful so we can exploit points 2 and 3. And after a while hopefully point 1 as well.

How the project functions

- https://www.capitoltrades.com/trades?page=1 is a website that displays trades of US politicians in compliance with the Stock Act of 2012 issued by Obama to combat insider trading. Politicians have to report their trades up to 45 days since making them. It contains 34520 individual trades dating back to June 2022.

Creating a Data-Sheet

Using a web-scrapper and a python library called yfinance I created 2 excels: One containing the politicians' names, the stocks they traded, amount of time they took to report the trade as well as how the stock performed in the 60 days prior to publishing of the trade. The second one contained the stocks performance 60 days into the future. Using the second excel I calculated the index of the maximum value of the stock in the future. I saved the index if the value was higher by 5% than when the report was published, or 0'ed it when it was not.

Jonathan Jackson	CRM.US	28	26	1	0.987711	0.975964	0.9625	0.98253	0.988042	0.989729	0.993193	0.970602	0.960753	0.938283	0.92244	0.941747	0.89259	0.891446	0.906627	0.873494	0.865121	0.868163	0.852952	0.834247	0.824458	0.848946	0.849548	0.828313
Tony Wied	TTD.US	4	25	1	1.044483	1.048653	1.0298	0.739314	0.717202	0.701129	0.692094	0.663467	0.658558	0.628063	0.64192	0.652129	0.634796	0.616334	0.616855	0.573154	0.580973	0.5649	0.566638	0.556386	0.526759	0.522937	0.500261	0.477932
Marjorie Taylor Greene	AMZN.US	3	26	1	0.991613	0.997505	0.991226	0.984301	0.985806	0.984172	0.969979	0.966796	0.960344	0.935269	0.910237	0.924473	0.93914	0.897419	0.917634	0.860688	0.88086	0.87914	0.858022	0.84129	0.833978	0.863312	0.852344	0.849075
Marjorie Taylor Greene	AAPL.US	4	26	1	0.988059	0.982162	0.995074	1.01965	1.038329	1.05081	1.053005	1.05421	1.058557	1.054167	1.06738	1.051585	1.030409	1.019822	1.040653	1.023093	1.013237	1.009019	1.011902	1.013753	0.963268	0.947472	0.929439	0.90921
Marjorie Taylor Greene	CAT.US	3	35	1	0.989384	0.979661	0.966825	0.954827	0.965741	0.962681	0.952932	0.961408	0.949303	0.929966	0.923494	0.935545	0.93067	0.926148	0.941449	0.886635	0.89132	0.907976	0.91916	0.934326	0.931401	0.928991	0.910034	0.914611
Marjorie Taylor Greene	DELL.US	3	36	1	0.998419	1.025298	1.023531	1.033203	1.032738	1.083612	1.105004	1.120722	1.123977	1.079892	1.055618	1.039807	1.08817	0.938988	0.963914	0.864955	0.880301	0.876581	0.853051	0.827753	0.842076	0.880394	0.873791	0.879836
Marjorie Taylor Greene	FDX.US	3	25	1	0.992971	0.998687	0.996601	1.012822	1.021047	1.038658	1.028539	1.029157	1.035259	0.982853	0.98351	0.991967	0.993203	1.003089	1.018074	0.986329	0.958098	0.95972	0.964123	0.974323	0.954636	0.945589	0.946482	0.939686
Marjorie Taylor Greene	FDX.US	4	25	1	0.992971	0.998687	0.996601	1.012822	1.021047	1.038658	1.028539	1.029157	1.035259	0.982853	0.98351	0.991967	0.993203	1.003089	1.018074	0.986329	0.958098	0.95972	0.964123	0.974323	0.954636	0.945589	0.946482	0.939686
Marjorie Taylor Greene	PI.US	3	26	1	0.979273	0.94981	0.968863	0.976392	0.962915	0.976206	0.974533	0.957338	0.943675	0.899154	0.877033	0.86746	0.897388	0.848592	0.902779	0.800539	0.847941	0.860303	0.85612	0.906311	0.935403	0.899433	0.851938	0.85138
Marjorie Taylor Greene	JPM.US	3	28	1	0.995961	0.974718	0.988495	0.993616	1.000361	0.998161	1.004003	1.005085	0.967721	0.957514	0.945757	0.927471	0.938363	0.940347	0.952141	0.920691	0.906156	0.894832	0.884445	0.851913	0.835828	0.843943	0.826992	0.822736
Marjorie Taylor Greene	LRCX.US	3	36	1	1.004327	0.998437	0.97704	0.980647	0.993509	1.020676	1.037384	1.08162	1.077533	1.039187	0.998437	0.981969	0.981488	0.922587	0.936892	0.913932	0.925492	0.929953	0.924166	0.92718	0.893179	0.908009	0.917655	0.925492
Marjorie Taylor Greene	LULU.US	3	30	1	0.961269	0.961438	0.948834	0.965881	0.95021	0.882986	0.893683	0.886681	0.883759	0.87893	0.884749	0.884556	0.880475	0.866084	0.883276	0.827305	0.830951	0.841937	0.838195	0.815111	0.805766	0.779905	0.763244	0.756749
Marjorie Taylor Greene	NKE.US	3	35	1	0.982993	1.000142	1.000567	1.026786	1.03699	1.055839	1.088719	1.089994	1.096939	1.117914	1.13818	1.163549	1.154904	1.123158	1.139383	1.106764	1.101494	1.09309	1.10363	1.104058	1.081267	1.059189	1.042381	1.043805
Marjorie Taylor Greene	NSC.US	3	37	1	0.997174	1.008359	0.999721	1.009872	1.026789	1.020619	1.022729	0.989451	0.990049	0.987183	0.974803	0.972693	0.961906	0.971897	0.982724	0.966722	0.955338	0.947019	0.948372	0.959518	0.948253	0.915492	0.92222	0.914059
Marjorie Taylor Greene	ODFL.US	3	25	1	1.014406	1.018274	1.024841	1.038432	1.053703	1.056605	1.055383	1.039247	1.00397	0.940901	0.919878	0.913209	0.895444	0.895393	0.906999	0.868618	0.894381	0.897032	0.905394	0.909982	0.904731	0.861903	0.84518	0.835901
Marjorie Taylor Greene	ODFL.US	4	25	1	1.014406	1.018274	1.024841	1.038432	1.053703	1.056605	1.055383	1.039247	1.00397	0.940901	0.919878	0.913209	0.895444	0.895393	0.906999	0.868618	0.894381	0.897032	0.905394	0.909982	0.904731	0.861903	0.84518	0.835901
Marjorie Taylor Greene	QCOM.US	3	24	1	0.999411	0.99894	0.998351	1.00212	1.01631	1.020197	1.024495	1.03633	1.030324	0.976918	0.951893	0.952894	0.955897	0.913266	0.936171	0.914267	0.912913	0.91233	0.918901	0.931511	0.916711	0.912922	0.903391	0.909074
Marjorie Taylor Greene	RH.US	3	25	1	0.967516	0.966814	0.907022	0.917036	0.931911	0.924776	0.923373	0.924389	0.89396	0.822606	0.834773	0.854437	0.849527	0.786324	0.781075	0.725637	0.706722	0.685074	0.671649	0.628885	0.568415	0.583678	0.572382	0.534577
Marjorie Taylor Greene	SO.US	4	0	1	1.013279	1.019739	1.022371	1.038282	1.041034	1.024289	1.03756	1.019463	1.056261	1.072307	1.076409	1.078339	1.064706	1.077736	1.081114	1.10271	1.076288	1.064706	1.068325	1.103313	1.115981	1.078822	1.075082	1.078098
Gil Cisneros	BCS.US	18	36	1	1	0.996047	1.016469	0.963109	0.973649	1.017787	1.013834	1.013175	1.019763	1.005929	1.019104	1.02108	1.025033	1.047274	1.072752	1.031854	1.062695	1.086162	1.064036	1.006376	0.982909	1.033865	1.029172	1.02716
Gil Cisneros	BSX.US	21	31	1	1.00926	1.010883	1.004678	1.011742	1.012888	0.989881	1.005728	1.006492	0.99895	0.995227	0.99284	0.958759	0.97031	0.965442	1.000955	0.998854	0.985585	0.986253	0.960477	0.933365	0.898616	0.935943	0.922578	0.917136
Gil Cisneros	CNC.US	7	0	1	0.999661	0.99746	0.971724	0.963766	0.96766	0.965967	0.949543	0.974771	0.959194	0.976634	0.997968	1.029292	0.975279	0.981713	0.98764	0.98764	0.982391	0.997291	1.001693	1.020826	1.010498	0.984931	0.98764	0.981036
Gil Cisneros	FERG.US	21	35	1	0.980528	0.989962	0.984696	0.999232	1.014755	1.019637	0.999616	0.983435	0.991279	0.954583	0.941802	0.97049	0.96292	0.962701	0.977675	0.939937	0.933959	0.917668	0.916077	0.905216	0.855027	0.866656	0.85393	0.860128
Gil Cisneros	MSFT.US	7	36	1	0.993349	0.983577	0.977742	0.977238	0.979135	0.979639	0.979351	0.999142	1.004074	0.98283	0.965002	0.957568	0.965411	0.944697	0.959517	0.922418	0.936709	0.948594	0.943879	0.928289	0.911832	0.921336	0.921841	0.913709

Creating the training set

Merging the 2 excels I got the training database, D column is the targeted amount of days after which it is best to sell.

POLITICIAN	TRADED ISSUER	PUBLISHED	TRADED	FILED AFTER	OWNER	TYPE	SIZE	PRICE
Markwayne Mullin Republican | Senate | OK	Adobe Inc ADBE:US	18:15 Yesterday	13 May 2025	29 days	Joint	BUY*	15K–50K	$397.40
Markwayne Mullin Republican | Senate | OK	Applied Materials Inc AMAT:US	18:15 Yesterday	13 May 2025	29 days	Joint	BUY*	15K–50K	$173.03
Markwayne Mullin Republican | Senate | OK	CSX Corp CSX:US	18:15 Yesterday	13 May 2025	29 days	Joint	BUY*	15K–50K	$30.54
Markwayne Mullin Republican | Senate | OK	Chevron Corp CVX:US	18:15 Yesterday	13 May 2025	29 days	Joint	BUY*	50K–100K	$142.35
Markwayne Mullin Republican | Senate | OK	ConocoPhillips COP:US	18:15 Yesterday	13 May 2025	29 days	Joint	BUY*	15K–50K	$94.17

Further usage

With the training set created all we have to do is provide new data to predict if the stock is worth buying. After the market closes I run the slightly modified webscraper to get all the trades that happened on that very day, then I run the script that gets me the past prices of the stock and then I pass the outcome into the model.

Profitability

I cut out a 500 trade sample from the training database, spanning 42 days and run 5 different models for it:

- Random Forest
- LGBM
- LSTM 200 epoch
- LSTM 400 epoch
- LSTM 600 epoch

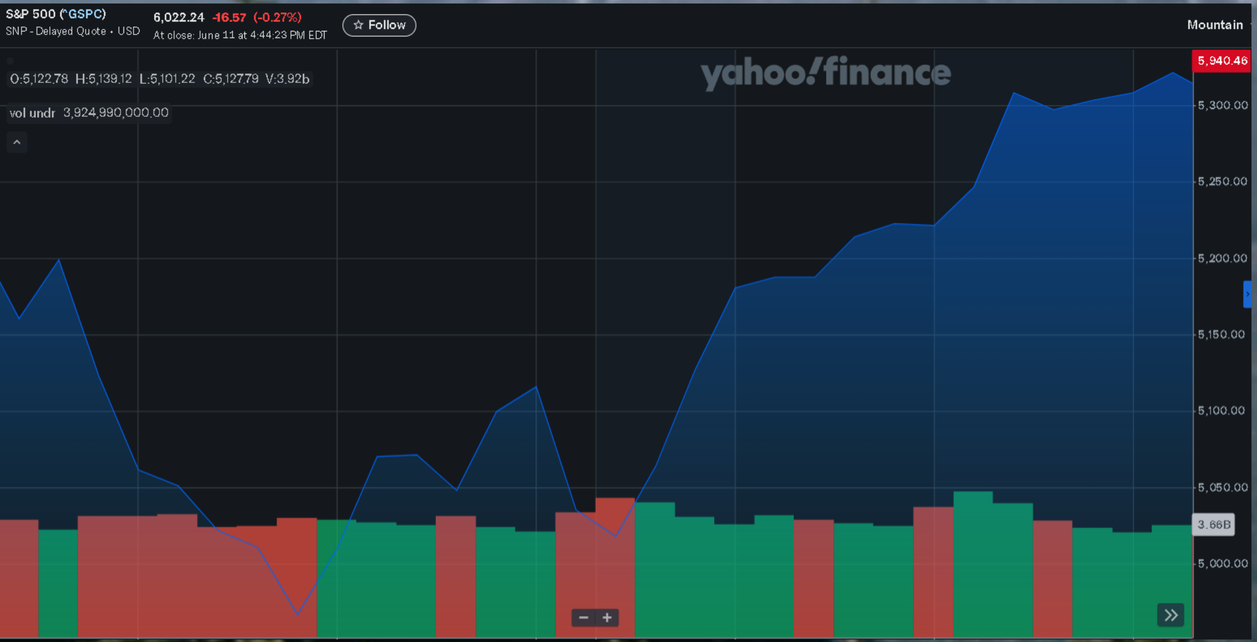
Profit

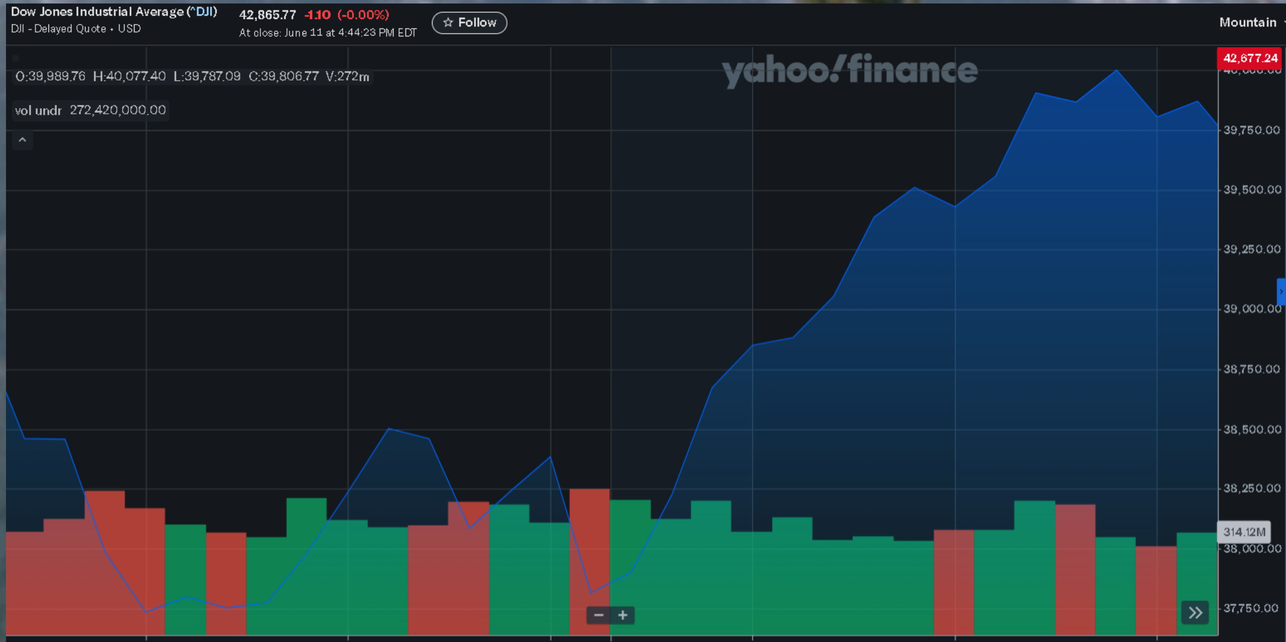
If one was to invest 1000 PLN into every trade the model made, he would get that much clean proft:

	RF	LGBM	LSTM 200	LSTM 400	LSTM 600
Profit	653.039	1491.4	-62.556	4811.9833	313.3388954
nr of trades	25	70	19	28	35
% Profit	2.61216	2.13057	-0.32924	17.185655	0.895253987
avr trade profi	3.0565	2.2667	-0.36798	18.507628	0.949511804

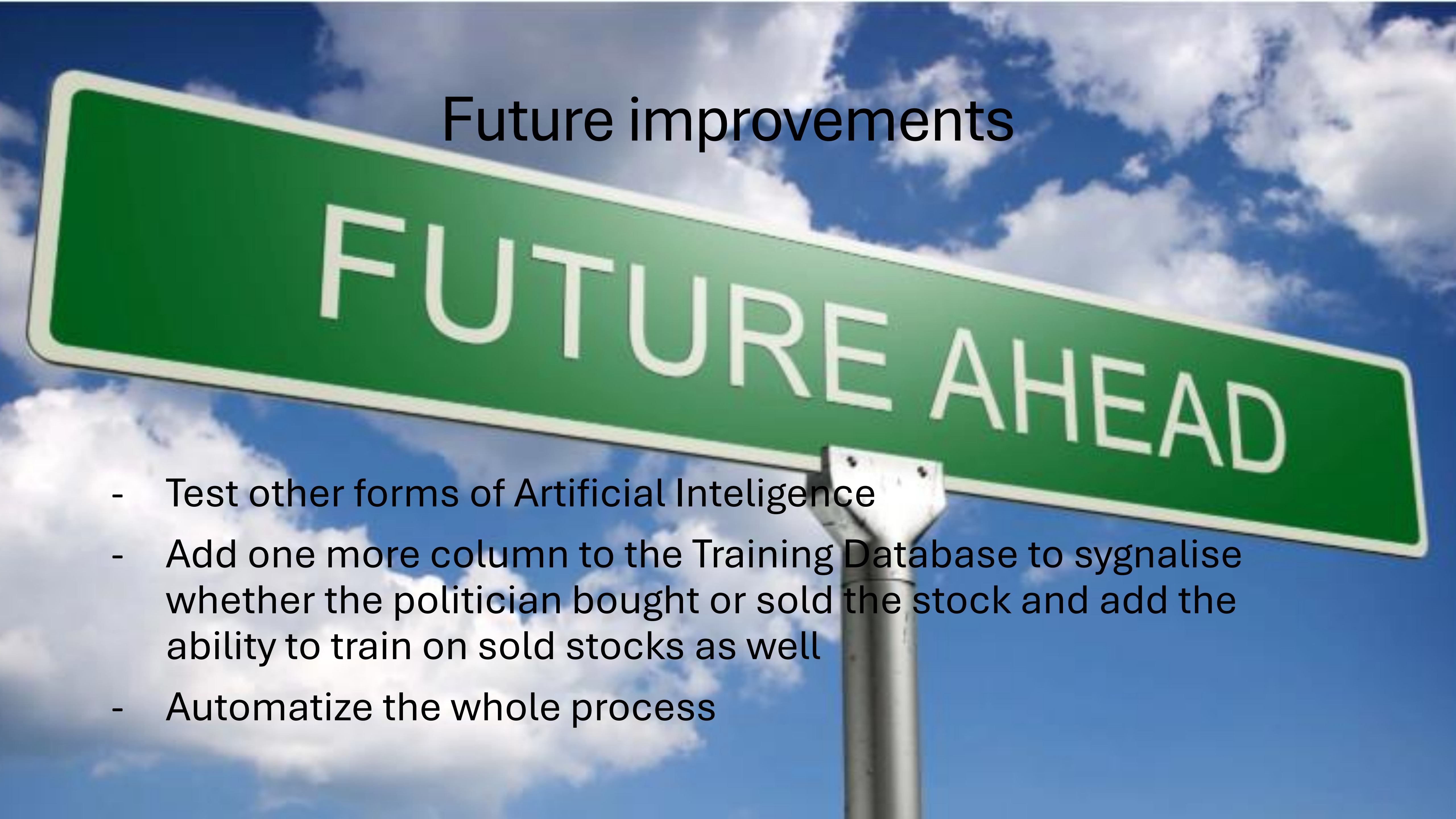
Comparison to other forms of investing

If we were to buy a safe index in the same time period as our sample it would generate us a profit o 7,83% for S&P500, 3,86% for Dow Jones





Future improvements



- Test other forms of Artificial Inteligence
- Add one more column to the Training Database to sygnalise whether the politician bought or sold the stock and add the ability to train on sold stocks as well
- Automatize the whole process

THANK YOU FOR YOUR ATTENTION